Cross-Lingual Transfer of Sentiment Classification: Multilingual versus Monolingual BERT for English and German

Samiya Mokhles Rahman
[redacted]
Heinrich Heine University Düsseldorf

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Abstract

Pretrained multilingual language models promise to reduce the need for labelled data in every language, but it is unclear how much performance is sacrificed relative to monolingual models, and how well such models transfer to a language with no training data at all. I compare monolingual BERT-family encoders against multilingual BERT (mBERT) on English/German sentiment classification of Amazon reviews, across five fine-tuning conditions plus a TF-IDF baseline, with the three conditions most central to my research question run over three seeds. Monolingual encoders outperform mBERT trained on the same language by 4.8–5.7 accuracy points, and mBERT fine-tuned only on English transfers to German with a further 7.4-point drop relative to mBERT fine-tuned directly on German. This zero-shot condition (76.1% mean accuracy) underperforms the TF-IDF baseline (81.8%) despite mBERT using roughly 60% more parameters, and shows the highest run-to-run variance of any condition. Error analysis reveals a strong bias: 88.5% of the zero-shot model's errors are positive German reviews misclassified as negative, concentrated in reviews with embedded negation words. These results suggest mBERT's shared cross-lingual representation space does not fully substitute for in-language supervision.

1 Introduction

English constitutes roughly half of all web content (49.6%, W3Techs, July 2026), while German, despite ranking among the ten most used languages online, accounts for only 5.9% (W3Techs, July 2026). Since labelled NLP datasets are typically built from web-sourced text, this imbalance is mirrored in the availability of labelled data for a given task: English benchmarks are abundant, while comparatively little labelled data exists for German and other languages.

Multilingual pretrained encoders such as mBERT are often proposed as a solution, since a single model is trained jointly across many languages and can transfer knowledge from data-rich to data-scarce languages without any target-language labels at all. What difference does this actually make? If multilingual models match monolingual performance, they offer an efficient way to extend NLP capabilities to lower-resource languages without collecting new labelled data. If they fall meaningfully short, one needs to weigh whether a language-specific model, or even a small amount of in-language labelled data, is worth the added cost.

In practice, however, it remains unclear whether multilingual mBERT matches monolingual encoders (BERT for English and a German-specific BERT for German) when each is fine-tuned in-language for sentiment classification, in terms of both performance and reliability. An additional question is whether mBERT can transfer zero-shot from English to German, without any German labels at all.

I address this through a controlled, five-condition comparison of monolingual and multilingual BERT-family encoders on a shared-domain English/German sentiment dataset, benchmarked against a non-neural TF-IDF baseline, isolating the cost of multilingual pretraining from the cost of lacking in-language supervision altogether.

2 Background and Related Work

BERT is pretrained on large unlabelled text corpora via masked language modelling, yielding contextual token representations that transfer well to downstream classification tasks once fine-tuned with a lightweight task-specific head. Multilingual BERT (mBERT) shares the same architecture, but is instead trained jointly on Wikipedia text from over 100 languages using a single shared subword vocabulary, giving a representation space that is only partially shared across languages. It was found that mBERT achieves non-trivial zero-shot cross-lingual transfer on tasks like named entity recognition and part of speech tagging, even though it receives no explicit cross lingual supervision during pretraining; transfer quality, however, varies considerably by task and language pair.

For the German monolingual condition, I use a German BERT model released by the MDZ Digital Library team (MDZ Digital Library Team (dbmdz), 2019), chosen over the more recent GBERT model since it uses the same WordPiece tokenization as English BERT and mBERT, keeping tokenization consistent across all three encoders. The dataset I used is the Multilingual Amazon Reviews Corpus (MARC) (Keung et al., 2020). All Transformer models were fine-tuned using the Hugging Face transformers library (Wolf et al., 2020); the baseline was implemented with scikit-learn (Pedregosa et al., 2011).

3 Data

3.1 Source and Licence

I use the Multilingual Amazon Reviews Corpus (MARC) (Keung et al., 2020), a collection of Amazon product reviews in six languages collected between November 2015 and November 2019, from which I only used the English and German subsets, accessed via the `SetFit/amazon_reviews_multi_{en,de}` mirror on the Hugging Face Hub. The corpus is distributed by Amazon under a non-commercial research licence. It may be analysed, and published on for academic purposes, but not resold or used commercially. This project uses the corpus strictly for non-commercial coursework, which is consistent with the licence terms.

3.2 Corpus Statistics and Label Construction

For each language, the original corpus provides 200,000 training, 5,000 validation, and 5,000 test reviews, so that each of the five star ratings (1–5) constitutes exactly 20% of the data. I binarised the star ratings into sentiment labels as follows: 1–2 stars being labelled negative, 4–5 stars being labelled positive, and 3-star reviews were dropped for being ambiguous. Due to time constraints, I use a smaller subsample of the full corpus that preserves the original proportion of each rating class (1–5), rather than the complete 200,000-example training sets.

The exact sizes are reported in Table 1.

Table 1: Dataset statistics by language and split, after binarisation and subsampling.

Language	Train	Validation	Test	% Positive
English	2,000	500	500	50.0%
German	2,000	500	500	50.0%

3.3 Preprocessing and Tokenization

BERT and the German BERT checkpoint both use WordPiece tokenization. mBERT uses the same WordPiece scheme, but over a larger shared multilingual vocabulary. All reviews are shortened or padded to a maximum sequence length of 128 subword tokens, which covers the large majority of reviews in this corpus given their typical length.

3.4 Language Balance and Dataset Quality

English and German subsets are drawn from the same corpus construction process, giving comparable review style, length distribution, and product-category coverage across languages. This is a strength of this dataset choice, since it isolates the effect of language without other domain differences getting in the way. I used equal example counts (2,000 train, 500 validation, 500 test) used for both languages, so that performance differences cannot be attributed to one language simply having more training data than the other.

3.5 Limitations and Ethical Considerations

Since the data comes from a single domain (product reviews) covering 2015 to 2019, results might not hold up for other forms of text, like social media posts or news articles, or for how language is used today. Star ratings also are not a reliable measure of sentiment. Someone could leave a 3-star rating while still writing something clearly positive or negative, and by simply dropping 3-star reviews to binarize the labels, I am ignoring that ambiguity rather than actually addressing it. Finally, since reviewer and product IDs are anonymised in the corpus, I am not able to check whether the same reviewer shows up in multiple splits.

4 Methodology and Experimental Setup

4.1 Baseline

As a non-Transformer baseline, I use TF-IDF features (unigrams and bigrams, vocabulary capped at 20,000 terms) combined with logistic regression, trained and evaluated separately per language using scikit-learn (Pedregosa et al., 2011). I use this baseline, because it is one of the requirements of this project to use at least one meaningful baseline, and this specific one lets me test whether the pretrained Transformer models actually earn their added complexity rather than just assuming bigger models are automatically better.

4.2 Models

- **Monolingual English**: `bert-base-uncased` (Devlin et al., 2019)
- **Monolingual German**: `dbmdz/bert-base-german-cased` (MDZ Digital Library Team (dbmdz), 2019)
- **Multilingual**: `bert-base-multilingual-cased` (mBERT) (Devlin et al., 2019)

All three checkpoints use WordPiece tokenization, which was a deliberate choice to keep tokenization consistent across every condition in the comparison.

4.3 Hyperparameter Summary

Table 2: Key hyperparameters for all conditions.

Transformer conditions (BERT, German BERT, mBERT)	
Maximum sequence length	128 tokens
Batch size	16
Epochs	up to 2, early stopping (patience 1) on validation macro-F1
Optimizer	AdamW
Learning rate	2×10^{-5}
Seeds	42 (all conditions); additionally 43, 44 for the three core German conditions
Baseline (TF-IDF + Logistic Regression)	
Vectorizer	TF-IDF, unigrams + bigrams, max 20,000 features
Classifier	Logistic Regression, max_iter=1000
Seed	42

4.4 Experimental Conditions

Table 3: The five experimental conditions, plus the baseline.

Condition	Trained on	Evaluated on
Baseline (TF-IDF + LogReg)	EN / DE separately	EN / DE respectively
Monolingual EN (BERT)	English	English
Monolingual DE (German BERT)	German	German
Multilingual EN (mBERT)	English	English
Multilingual DE (mBERT)	German	German
Zero-shot transfer (mBERT)	English only	German

For the zero-shot transfer condition, early stopping used the English validation set and not the German. German labels were not used during training or model selection for this condition. The German test set was only used for the final evaluation.

4.5 Controlled Variables

Across all five Transformer conditions, only the model checkpoint and training languages vary. Every other setting (Table 2) is held fixed, including the evaluation code and metric implementation.

4.6 Fine-Tuning Procedure

All Transformer conditions use full fine-tuning (every parameter updated), following the hyperparameters in Table 2, with the best-validation-F1 checkpoint restored before final test evaluation. Given time constraints, I prioritised random-seed coverage for the three conditions most relevant to the research question, which were: monolingual German, multilingual in-language German, and multilingual zero-shot transfer. I ran each of them with three seeds. The remaining conditions were run with a single seed.

4.7 Evaluation Metrics

I looked at both accuracy and macro-F1 on the test set. My binarised labels came out exactly balanced (50% positive / 50% negative in every split, Table 1), so I expected the two metrics to land close to each other and that's exactly what happens in my results (Section 5). I kept macro-F1 in a way to confirm the model isn't favoring one class over the other.

4.8 Reproducibility

All experiments were run on a single Kaggle T4 GPU. Software versions: `transformers` ≥4.35, `datasets` ≥2.14, `torch` ≥2.0, `scikit-learn` ≥1.3. Seed 42 was used for all conditions; seeds 43 and 44 were additionally used for the three conditions central to the research question (Section 4). Total wall-clock training time for the single-seed pass across all five Transformer conditions was approximately 8–9 minutes across repeated runs (Table 5), plus a further ∼10 minutes for the six additional reseeded runs, excluding one-time model/tokenizer downloads. I independently reran the full pipeline twice and all reported metrics (accuracy, macro-F1, and their per-seed standard deviations) matched exactly between runs, supporting the reproducibility of the reported numbers.

5 Results

Table 4 reports test-set accuracy and macro-F1 for all conditions. As described in Section 4, the three conditions most central to the research question were each run with three seeds and are reported as mean ± std; the remaining four conditions (both baselines, monolingual English, and multilingual in-language English) use a single seed due to time constraints. Validation-set macro-F1 was used only for early-stopping/model selection during training and is not reported separately here. It is also consistent with the standard practice of reporting final performance on the held-out test set.

Table 4: Test-set accuracy and macro-F1 by condition. Three conditions central to the research question (marked †) report mean ± std over 3 seeds (42, 43, 44). The remaining conditions report a single seed (42) due to time constraints.

Condition	Accuracy	Macro-F1
Baseline (TF-IDF+LogReg), EN	0.874	0.874
Baseline (TF-IDF+LogReg), DE	0.818	0.818
Monolingual EN (BERT)	0.920	0.920
Monolingual DE (German BERT)†	0.893 ± 0.006	0.893 ± 0.006
Multilingual EN (mBERT)	0.872	0.872
Multilingual DE (mBERT)†	0.835 ± 0.017	0.835 ± 0.017
Zero-shot transfer (mBERT)†	0.761 ± 0.027	0.751 ± 0.031

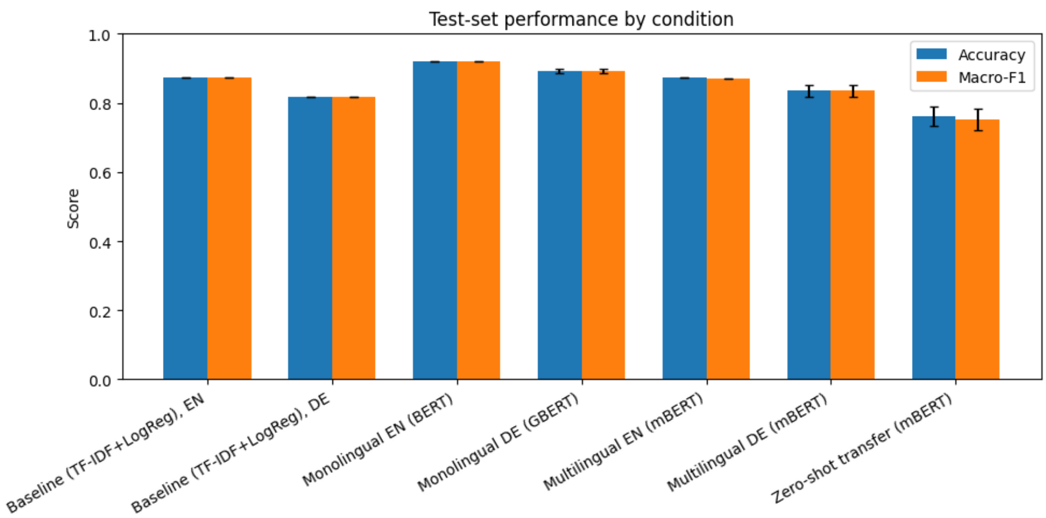

Figure 1: Test-set accuracy and macro-F1 across all seven conditions. Error bars show ±1 standard deviation for the three conditions run with 3 seeds (monolingual DE, in-language multilingual DE, and zero-shot transfer), the remaining conditions use a single seed and have no error bars.

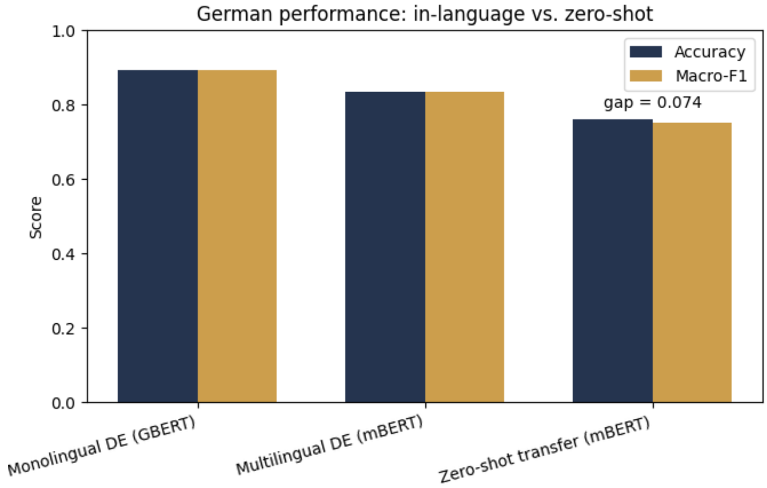

Figure 2: German test performance: monolingual in-language training, multilingual in-language training, and multilingual zero-shot transfer from English. The annotated gap is the accuracy difference between multilingual in-language mBERT and zero-shot transfer.

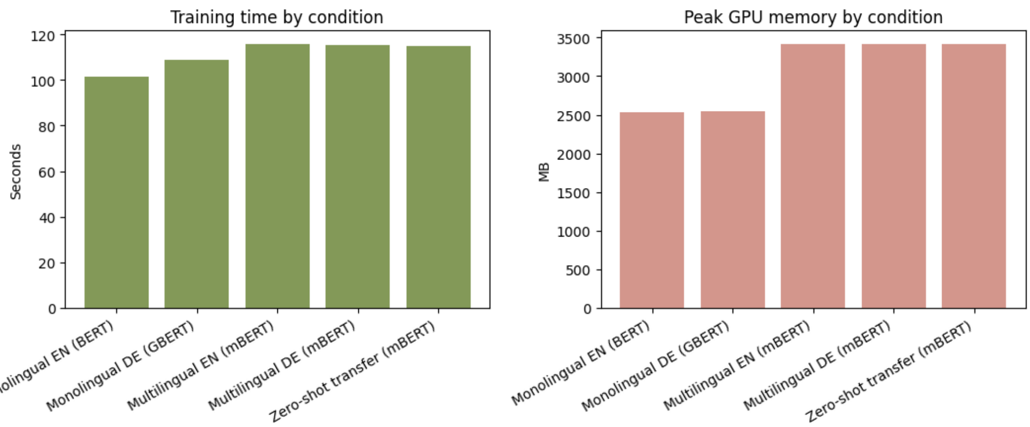

Figure 3: Training time and peak GPU memory by condition.

6 Discussion, Error Analysis, and Limitations

6.1 Do the Transformer models justify their complexity?

All Transformer conditions outperform the TF-IDF baseline in their respective in-language setting. For example: monolingual BERT beats the English baseline by 4.6 accuracy points (92.0% vs. 87.4%), and in-language mBERT and monolingual German BERT both beat the German baseline (83.5% and 89.3% mean accuracy vs. 81.8%). This confirms that pretrained contextual representations provide a real benefit over simple word-frequency statistics for this task, in both languages. However, this advantage disappears in the zero-shot transfer condition.

6.2 Monolingual versus multilingual: an in-language cost

Comparing multilingual mBERT against the corresponding monolingual model in the same language reveals a consistent gap: mBERT underperforms BERT by 4.8 points on English (87.2% vs. 92.0%, single seed) and underperforms the monolingual German model by 5.7 points on German (83.5% vs. 89.3% mean accuracy over 3 seeds). This suggests mBERT pays a real cost for being spread across so many languages at once. Its fixed capacity has to be shared, so it loses some language-specific precision, even after it's fine-tuned on labelled data for just one target language.

Table 5: Computational cost by condition.

Condition	Train time (s)	Peak memory (MB)	Trainable params
Monolingual EN (BERT)	101.5	2,532	109,483,778
Monolingual DE (German BERT)	108.7	2,546	109,929,218
Multilingual EN (mBERT)	115.9	3,419	177,854,978
Multilingual DE (mBERT)	115.5	3,419	177,854,978
Zero-shot transfer (mBERT)	115.1	3,419	177,854,978

6.3 The zero-shot transfer gap

The comparison between mBERT fine-tuned directly in German (83.5% $\pm$ 1.7 mean accuracy) and mBERT fine-tuned only in English evaluated zero-shot in German (76.1% $\pm$ 2.7 mean accuracy) which is a gap of 7.4 accuracy points (Table 4, Figure 2; note Figure 2 shows the original single-seed run, while the text here reports the updated 3-seed means). This shows that mBERT's shared cross-lingual representation space supports substantial, but incomplete, transfer. A large share of the in-language performance gain from fine-tuning on German specifically is lost when no German labels are available at all, but the model still performs far above chance (50%), indicating it is not merely guessing.

More strikingly, zero-shot transfer (76.1% mean accuracy) underperforms the simple German TF-IDF baseline (81.8% accuracy) by 5.7 points, despite mBERT having roughly 60% more parameters than the monolingual models (Table 5) and having been exposed to substantially more pretraining data overall. This is a meaningful finding: for a practitioner with no labelled German data but the ability to obtain a small amount of labelled German data cheaply (e.g. via crowdsourcing), training a trivial baseline directly on that small labelled set may be preferable to relying on zero-shot transfer from a large multilingual model trained only on English.

A further finding only visible with multiple seeds: zero-shot transfer has the highest variance of the three reseeded conditions (accuracy std 2.7 points, vs. 0.6 for monolingual German and 1.7 for in-language mBERT). Zero-shot transfer is therefore not only weaker on average but also less reliable across random initialisations – which matters in practice, since a single lucky run could otherwise make this transfer strategy look more dependable than it actually is.

6.4 Error analysis

I extracted every misclassified example from the zero-shot condition (seed 42) on the German test set: 104 of 500 examples (20.8%) were misclassified, consistent with that seed's 79.2% accuracy. Critically, these errors are not evenly split between the two error types: 92 of the 104 errors (88.5%) were false negatives – positive reviews predicted as negative – while only 12 (11.5%) were false positives which is a strong, systematic bias and not just noise. A model with no directional bias would be expected to split errors far more evenly between the two classes given the balanced test set.

Manual inspection of the false-negative examples reveals two recurring linguistic patterns. One pattern involves reviews that start with mild criticism but end up positive:

> "Anfangs etwas ungewohnt in der Bedienung. Wir haben uns aneinander gewöhnt und ich bin mit der Leistung zufrieden." ("A bit unfamiliar to operate at first. We got used to each other and I am satisfied with the performance.")

Second, negation words embedded in ultimately positive sentences, where a negation particle (nicht) or a negative-sentiment root (negativ) appears without negating the review's overall sentiment:

> "Kann die negativen Bewertungen nicht verstehen." ("Can't understand the negative reviews" – the reviewer disagrees with other, more critical reviews.)

This suggests the model learned a surface-level link between negation or criticism of words and negative sentiment from English, but never learned to properly handle these structures in German which is likely because it saw no German training data. This explains the zero-shot gap in Section 5: the model isn't just noisier on German, it's systematically biased toward negative predictions.

6.5 Computational cost

mBERT uses approximately 62% more trainable parameters than either monolingual model (177.9M vs. 109.5–109.9M) and consumes roughly 35% more peak GPU memory (3,419MB vs. 2,532–2,546MB), for training time only marginally longer (100s vs. 93s for a fixed dataset size and epoch budget; Figure 3). Combined with its in-language performance deficit relative to monolingual models (Section 6), this suggests that when in-language labelled data is available in sufficient quantity, a monolingual model is the more efficient choice on every axis measured here. Multilingual pretraining's advantage is specifically in enabling transfer to languages with no labelled data, not in matching or exceeding monolingual performance where labels exist.

6.6 Limitations

The main limitations of this study are: (1) three seeds were run only for the conditions most central to the research question (monolingual German, in-language multilingual German, and zero-shot transfer); the remaining four conditions use a single seed, so the English-side comparisons should be read with a bit more caution than the German-side ones; (2) a reduced training set size (2,000 examples per language rather than the full 200,000 available) due to time constraints; (3) a single dataset and domain (Amazon product reviews), so findings may not generalise elsewhere; (4) the error analysis covers only the single seed-42 zero-shot run and a subset of its 104 misclassified examples, read manually rather than systematically annotated; and (5) only one German monolingual checkpoint was tested.

6.7 Possible improvements

In the future one could extend multi-seed coverage to the remaining conditions by scaling up to the full 200,000-example training sets, extend the error analysis with systematic linguistic annotation across misclassified examples pooled from all three zero-shot seeds to test whether the false-negative bias found here is stable across runs and compare a second German checkpoint (e.g. GBERT) and test transfer to a more distant target language to see whether the gap widens further.

7 Conclusion

This project compared monolingual and multilingual BERT-family encoders on English/German sentiment classification. Monolingual encoders consistently outperformed multilingual mBERT trained on the same language by 4.8–5.7 accuracy points (single-seed on English, 3-seed mean on German), confirming a real in-language cost to multilingual pretraining. mBERT's zero-shot transfer from English to German reached 76.1% mean accuracy over three seeds well beyond what one would expect from random guessing alone, but 7.4 points below mBERT trained directly on German, and, notably, 5.7 points below a simple TF-IDF baseline trained directly on German. Zero-shot transfer also showed the highest run-to-run variance of any reseeded condition, indicating it is both weaker and less reliable than in-language training. I conclude that mBERT's shared cross-lingual representation space enables substantial but incomplete transfer for this task and domain, and does not straightforwardly substitute for even a small amount of in-language supervision. A natural next step would be to extend multi-seed coverage to the remaining conditions, scale up to the full-size dataset, and test transfer to a linguistically more distant target language.

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Declaration of AI Usage

Claude (Anthropic) was used in this project to brainstorm and narrow down the project topic and research question and identify candidate datasets and their licensing terms while helping with debugging environment and dependency errors encountered on Kaggle (tokenizer and compatibility issues). Moreover it was used to draft the initial code structure for data loading, model training, and result visualisation. All code was executed and all experiments were run by me on Kaggle and all reported numerical results are taken directly from that execution and were not altered or fabricated. I reviewed, adjusted, and am able to explain all code and methodological decisions in this report, including the substitution of `dbmdz/bert-base-german-cased` for `deepset/gbert-base` (made to resolve a tokenizer dependency error and to keep tokenization consistent across conditions, as described earlier.)